

Technical and Commercial Proposal

Provider-Agnostic Wallet Base Architecture

Polity Wallet Infrastructure Service - R1.3.2 Base Layer

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1. Introduction

1.1 Background and Context

Polity is procuring the provider-agnostic wallet base architecture defined in RFP-T-BACK-0006 V.2.3.1. The work sits at the base of the R1.3.2 wallet infrastructure stack: it establishes the shared contract, persistence, orchestration, and evidence layer that the downstream Fireblocks, DFNS, and Safeheron provider RFPs will consume after acceptance. The RFP is explicit that this engagement does not activate a Listed Wallet Provider, enable production signing, or move sensitive operations into live use.

Turntabl has reviewed RFP-T-BACK-0006 V.2.3.1 in full, including all 26 WBS-BASE rows, the six MR-BASE review packages, Appendix A technical guidance, Appendix B FR/NFR traceability, and Appendix C glossary. We submit this proposal as one fixed-price package covering WBS-BASE-01 through WBS-BASE-26, with transparent role/rate/hour assumptions, approval-only contingency, and milestone-based payment under the six-milestone schedule in RFP §12.2.

1.2 Proposed Approach

Turntabl proposes a pure-backend Go implementation inside the existing wallet-infrastructure-service (WIS) module, using the direct Temporal Go SDK posture, temporal-saga-go (github.com/sunny-b/temporal-saga-go) for SAGA compensation, gormigrate for reversible schema migrations, and the Polity Vault REST credential gateway for all provider secret access at Activity execution time. The work is sequenced as a controlled Strangler Fig transition: existing DFNS and Safeheron read behaviour remains reachable through compatibility wrappers until the unified API, persistence, and evidence path are accepted. HTTP 301 redirection of legacy endpoints is deferred to R1.3.3 per the wallet implementation baseline.

The Base RFP delivery is implementation-focused and within guardrails. We will not redesign architecture, introduce alternative SAGA frameworks, modify Dynamic.xyz or Veriff integrations, deploy unmanaged Temporal infrastructure, activate Nexus runtime routing, or expose sensitive operations (signing, payment, raw broadcast, WalletConnect, transaction execution) without explicit governance approval. Fireblocks is represented at the architecture identity, registry slot, capability descriptor, and disabled-fixture level only, in line with RFP §4 current-state evidence.

1.3 Acceptance of Commercial and Methodological Terms

Turntabl accepts the RFP's commercial and methodological constraints:

- AI-assisted coding is our default delivery mode and matches the RFP Effort/Budget Baseline assumption. We accept the 67-hour planning baseline.
- Pricing is quoted in EUR per the RFP Quotation Currency metadata row.
- Contingency is shown separately, capped at fifteen percent (15%) of fixed price, and is approval-only (not pre-approved).
- We accept the six-milestone payment schedule in RFP §12.2 (10% / 10% / 20% / 30% / 25% / 5% warranty release at 90 days).
- We accept the service credit clause: 0.5% per full week of vendor-caused delay, capped at 5% of fixed price, as a deduction from the next due milestone payment.
- We accept the five percent (5%) warranty holdback released 90 calendar days after Final Acceptance subject to no material defect raised by Polity.
- Prompt-hygiene controls in the Confidentiality/NDA row and the AI-assisted coding controls row of the Risk Register are accepted: no Polity Confidential Information, secrets, Vault references, workflow inputs/history, signing payloads, raw provider keys, or live database extracts will be transmitted to any third-party AI provider.

1.4 Key Benefits to Polity

- Strict alignment to the binding MR-BASE-01.06 packaging and the WBS-BASE-01.26 task structure with no remapping or scope drift.
- Senior Go team familiar with deterministic workflow patterns, SAGA compensation, and provider-neutral adapter abstractions.
- Polity Vault REST gateway boundary respected end-to-end: WIS does not call Infisical directly; credentials are fetched only at Activity execution time and never serialised into workflow history, logs, fixtures, frontend code, provider metadata, or ordinary wallet rows.
- Sensitive-operation closure delivered as a three-part evidence pack: static code scan with admin-scope analysis, runtime API inventory, and custodian attestation with supporting Listed Wallet Provider evidence where applicable.
- Full evidence package, CSV-ready traceability index across all 26 WBS rows, runbook, and provider-start handoff that enables the Fireblocks, DFNS, and Safeheron provider RFPs to begin without re-reading the base codebase.

2. About Turntabl

2.1 Company Overview

Turntabl is a technology solutions provider focused on the design, development, and delivery of custom backend systems for enterprise and financial-sector organisations. The company specialises in secure, scalable services delivered under structured governance with documented acceptance evidence. Turntabl operates with a strong emphasis on quality, reliability, and accountability, and has delivered for tier-1 global investment banks, asset management firms, and fintechs operating under stringent regulatory and operational requirements.

2.2 Relevant Experience

Turntabl brings directly relevant experience in the capabilities required for this engagement:

- Go backend services with GORM and gormigrate-based reversible migrations, structured error taxonomies, idempotency-keyed operation ledgers, and provider-neutral abstraction layers.
- SDK orchestration with deterministic workflow logic, bounded activity retry policies, `NonRetryableErrorTypes`, heartbeat for long-running activities, and SAGA compensation patterns including the temporal-saga-go family of helpers.
- Multi-provider integration architectures where provider-specific SDKs are wrapped behind adapter interfaces, capability metadata gates execution, and credential access is brokered through centralised secret-management gateways at execution time.
- Security engineering aligned with OWASP API Security Top 10 and Zero Trust principles: fail-closed webhook envelopes with HMAC verification, route-fate inventories, sensitive-operation negative-capability proofs, and PII-minimised structured audit trails.
- Governance-ready delivery within structured CSV evidence frameworks, MR-based workflows with milestone-gated acceptance, traceability matrices, runbook documentation, and DNO-gated handling of procurement and technical evidence.

2.3 Delivery Philosophy

- Implementation within guardrails: Polity's architectural decisions and baseline authority are authoritative. Conflicts with the Polity system requirements baseline, Polity system architecture baseline, Polity wallet architecture baseline (ADR-T-BACK-WIP, ADR-GLB-WIP), or Polity wallet implementation baseline (MIG-T-BACK-WIP, MIG-GLB-WIP) are surfaced as written exceptions under PDK §11.1, not resolved unilaterally.
- Evidence-led, reversible delivery: every MR-BASE package is self-contained and ships with rollback evidence. No package depends on a future package to be safe. Every WBS row maps to a deliverable, a test/evidence artifact, an MR-BASE acceptance gate, and a rollback path.
- Transparent estimation: where we believe a WBS row requires more effort than the 67-hour baseline allows, we say so in this proposal rather than discovering hidden work later. Where we accept the baseline, we accept the AI-assisted coding productivity assumption it depends on.

3. Understanding Polity's Requirements

3.1 Core Problem

Per RFP §4 Current System Overview, the existing WIS codebase exposes provider-specific routes for DFNS and Safeheron at `internal/rest/router/router.go` and `internal/rest/api/api.go`; `internal/service/interface.go` mixes payment, broadcast, and personal-sign methods into the wallet service surface; provider credentials in `polity_vault/wallet_credentials/models.py` support DFNS and Safeheron schemas only; the frontend type at `frontend/src/apps/cabinet/features/Wallet/types/base.ts` permits safeheron | dfns; and `internal/adapters/dfns`, `internal/adapters/safeheron`, and `internal/adapters/fireblocks` directories exist but are empty. There is no repository-evidenced Fireblocks runtime, no repository-evidenced Temporal SDK wallet orchestration path, and no provider-neutral persistence model.

This creates four compounding problems that the Base RFP must close before any provider-specific RFP can begin: (1) there is no canonical operation contract or registry that all providers consume; (2) sensitive operations are reachable through legacy paths without static or runtime negative-capability proof; (3) operation idempotency and replay safety are not enforced by a provider-neutral ledger; (4) credential access patterns do not yet route exclusively through the Polity Vault REST gateway.

3.2 What Must Be Delivered

The base layer comprises the following deliverable groups, each traced to its MR-BASE package below and to its Polity baseline authority per RFP §6.1:

- Current-state inventory and base contract freeze (MR-BASE-01): route inventory, sensitive-path register, persistence and credential inventory, base contract freeze note, blocked-decision log.
- Provider-neutral contract (MR-BASE-02): ProviderAdapter interface with disabled sensitive operations, canonical errors and capability metadata, AdapterRegistry with conformance fixtures, feature gates and sensitive-operation kill switch, Vault gateway request contract.
- Provider-neutral persistence (MR-BASE-03): reversible gormigrate steps for providers, `provider_accounts`, `wallets`, `wallet_assets`, `wallet_operations`, and `provider_metadata`; seed data and legacy DFNS/Safeheron read compatibility; WALLET_OPERATIONS ledger with deterministic idempotency; rollback package.
- Direct Temporal orchestration (MR-BASE-04): worker bootstrap, direct client.ExecuteWorkflow boundary, Tier 1 ProvisionWalletWorkflow and RegisterAccountWorkflow, ten Tier 3 Canonical Activities, retry and SAGA compensation evidence, workflow determinism and history hygiene proof.
- Unified API and route fate (MR-BASE-05): `/api/wallets/v1/*` skeleton or compatibility bridge preserving DFNS/Safeheron read behaviour; route fate matrix classifying every wallet route as preserve, wrap, deprecate, disable, or remove; sensitive-operation negative-capability closure; Dynamic/Veriff/Lukka non-regression; callback/webhook envelope boundary with fail-closed signature verification result handling.
- Traceability and handoff (MR-BASE-06): CSV-ready traceability matrix across all 26 WBS rows mapped to FR/NFR/TOR gates/MR-BASE/test/rollback/owner; runbook and provider-start handoff checklist; final QA and release-boundary review.

3.3 Key Architectural Constraints (Acknowledged and Accepted)

We accept and will comply with the following non-negotiable constraints from RFP §5 and §7-§10:

- The WIS Go module is the sole implementation target. We will not create parallel EKS resources, deploy an unmanaged Temporal Server, bypass Helm/GitOps, or store provider secrets in Helm values. Where infrastructure-delivery baseline handoff is incomplete, we record the blocker and use local Docker Compose Temporal or mock proof.

- Credential access is executable only through the Polity Vault REST gateway at service or Activity execution time. WIS will not call Infisical directly, will not store raw provider secrets, and will not serialise credentials into workflow inputs, workflow history, logs, fixtures, frontend code, provider metadata, or ordinary wallet rows.
- The AdapterRegistry rejects duplicate provider registration at startup, returns a controlled unknown-provider error when a provider id is not registered, and exposes capability metadata that separates implemented, disabled, unsupported, sandbox-only, and deferred states. Capability metadata alone is not authorisation.
- Workflow code is strictly deterministic: no secrets, no SDK clients, no file I/O, no database writes, no direct Vault calls, no provider clients, no non-deterministic side effects in workflow state. All external calls and persistence writes belong inside Activities.
- Direct Temporal client.ExecuteWorkflow is the R1.3.2 invocation path. Nexus routing remains disabled; wallet.nexus.routing.enabled remains false in all environments. The Nexus Service contract is registered on the Worker for validation purposes only.
- Tier 1 base workflows are ProvisionWalletWorkflow and RegisterAccountWorkflow. Tier 2 Provider-Specific Procedures are out of base scope. Tier 3 Canonical Activities are the ten enumerated in RFP §20.1.9 and §11 (WBS-BASE-17).
- SAGA compensation uses github.com/sunny-b/temporal-saga-go (the binding framework named in RFP §20.1.10) with forward-recovery, step-status tracking, per-Activity compensation registration, and reverse-order compensation on downstream failure.
- Route migration is a controlled Strangler Fig per RFP §7. Existing DFNS and Safeheron routes remain reachable as compatibility wrappers in R1.3.2. HTTP 301 redirection of legacy endpoints is deferred to R1.3.3 and is not part of this engagement.
- Sensitive operations (signing, payment, raw broadcast, WalletConnect, transaction execution, scan, provider dashboard, sensitive admin paths) are blocked by both static capability closure and runtime feature gates. The seven-step gate precedence in RFP §9 must pass before any provider side effect can occur.
- Fireblocks is represented at the provider identity, registry slot, capability descriptor, and disabled-fixture level only. Fireblocks SDK lifecycle flow, webhook authentication, sandbox proof, production accounts, wrapper-wallet persistence, transaction/signing/broadcast/payment/WalletConnect behaviour, institutional vault, and Polity-held workspace-key patterns remain provider-RFP scope.

3.4 Observations and Clarification Requests

The following items require confirmation at engagement start. Defaults below reflect our reading of the RFP and current WIS evidence; where Polity confirms a different position, we will adjust the affected WBS rows before implementation begins. Items will be re-validated during the WBS-BASE-01 current-state inventory.

- AC-01-AC-10 (excluding AC-06) and G0-G7 TOR section references: supplied by Polity at engagement start under PDK §11.1 (per RFP §6.2). We treat the schedules as binding once supplied and will not infer meaning from the codes alone.
- Parent TOR identifier (Wallet Infrastructure Base Enablement TOR) and PDK identifiers for the Polity baselines listed in RFP §6.1: supplied through the engagement pack under controlled disclosure.
- HTTP router library: RFP §4 evidence indicates the existing WIS uses fasthttp/fasthttp/router. We will reuse fasthttp idioms in our new handlers unless WBS-BASE-01 inventory shows otherwise. Default: fasthttp.
- Existing internal/contract and internal/repository package locations: we will place the provider-neutral contract package and the WalletRepository at the locations specified in RFP §20.2, or the closest existing local equivalents identified during WBS-BASE-01.
- Worker endpoint: we will use wallet-infrastructure-service/cmd/worker/main.go per RFP §11 (WBS-BASE-15), or the existing equivalent identified at WBS-BASE-01.

- Provider sandboxes (DFNS, Safeheron) and mock-adapter availability: required for evidence production. Where any sandbox is unavailable, we will produce fixture-based evidence and a blocked-evidence register entry pending provider-RFP closure.
- polity_tests integration evidence repository identifier and access path: supplied by Polity at engagement start (per RFP §22.1 Glossary).
- Temporal connection details (namespace polity-wallet, task queue wallet-operations, and proof-lane connection string): supplied by Polity through the infrastructure delivery baseline handoff. We will use local Docker Compose Temporal until the EKS/proof-lane handoff is available, recording the dependency in the blocked-condition register.

4. Technical Approach

This section maps every WBS-BASE row to its MR-BASE package and concrete implementation approach, using the binding MR-BASE-01-06 grouping defined in RFP §11 and §14. Each package below states its purpose, WBS rows, hour allocation per RFP §11, our implementation approach, deliverables, acceptance gate, and stop conditions. The Polity baseline traceability for each WBS row follows Appendix B (BASE-FR/BASE-NFR) and the Parent requirement trace column.

4.1 MR-BASE-01 - Current-State Inventory and Base Contract Freeze (6 hours)

MR-BASE-01 covers WBS-BASE-01 through WBS-BASE-03. The package produces the route inventory, sensitive-path register, persistence and credential inventory, and base contract freeze note before any code changes are made. The acceptance gate is contract freeze, route/table inventory, sensitive-path inventory, and exception log accepted by Polity. Runtime provider implementation, schema changes, and route activation are explicitly out of scope for this package.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-01	2h	Inventory wallet routes in internal/rest/router/router.go and internal/rest/api/api.go, including DFNS/Safeheron legacy paths and transaction, signing, raw broadcast, payment, WalletConnect, scan, webhook, callback, and admin surfaces. No code activation.	Route inventory; sensitive-path register; exception log. Static route scan, endpoint list, reviewer sign-off on sensitive paths.
BASE-02	2h	Inventory wallet tables, gormigrate history, provider metadata, existing credential access via providers/polityVault, and the Polity Vault REST gateway. Direct WIS Infisical access, secret storage, private keys, and signing payload storage are excluded.	Table inventory; migration register; credential boundary note proving existing secrets remain behind the Vault gateway.
BASE-03	2h	Freeze the base contract artefact: ProviderAdapter, ProviderCredentials, WalletOperationsService, canonical errors, and ProviderCapabilities at the proposed wallet-infrastructure-service/internal/contract package (or closest existing local equivalent).	Contract freeze note; provider handoff assumptions; blocked-decision log. Reviewed contract notes showing Fireblocks, DFNS, and Safeheron provider RFPs consume the same base artefact without local forks.

Stop condition (RFP §14.1): Stop if code changes are needed before current-state evidence is accepted.
Branch guidance: feature/wallet-base/base-01-inventory-contract-freeze.

4.2 MR-BASE-02 - Provider-Neutral Contract, Registry, Capability, Gates, Vault Boundary (13 hours)

MR-BASE-02 covers WBS-BASE-04 through WBS-BASE-08. The package implements the ProviderAdapter interface with disabled sensitive operations, canonical errors and capability metadata, the AdapterRegistry with conformance fixtures, the feature-gate precedence chain with sensitive-operation kill switch, and the Polity Vault REST gateway request contract. The acceptance gate is interface tests, registry tests, disabled-operation tests, gate matrix, and no-secret Vault-boundary tests all passing. Provider API client logic, provider SDK

activation, production accounts, direct Infisical calls, signing, raw broadcast, and payment execution are explicitly out of scope.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-04	3h	ProviderAdapter and ProviderCredentials interfaces for lifecycle and visibility operations only. Unsupported transaction execution, direct provider signing, raw broadcast, payment, and WalletConnect invocation return canonical PROVIDER_VALIDATION disabled/unsupported responses. Method surface follows RFP §20.1.6: Identity, Account Lifecycle, Wallet Lifecycle, Asset Lifecycle, Transaction Scaffolding (deferred), Signing Discovery, Metadata, Reconciliation.	Interface types; disabled-operation behaviour; compile proof. Unit tests for supported and unsupported operations plus static proof that sensitive-operation methods cannot reach provider runtime paths.
BASE-05	2h	Canonical error classes per RFP §20.1.5: PROVIDER_TIMEOUT, PROVIDER_RATE_LIMIT, PROVIDER_UNAVAILABLE (retryable); PROVIDER_CONFLICT (CONFLICT_IDEMPOTENT), PROVIDER_AUTH_FAILURE, PROVIDER_NOT_FOUND, PROVIDER_VALIDATION (non-retryable); PROVIDER_INTERNAL (retryable, 3-cap then compensate); PROVIDER_PENDING (polling status). Capability flags, supported chain metadata, and display-safe provider metadata.	Error taxonomy; capability metadata model; fixture matrix. Table-driven tests prove retryable, non-retryable, unsupported, conflict-as-idempotent-success, and deferred capability handling.
BASE-06	3h	AdapterRegistry duplicate-provider rejection at startup, unknown-provider controlled error, DFNS/Safeheron wrapper fixtures, and disabled Fireblocks Embedded Wallet / NCW fixture. Conformance fixture suite that every provider adapter must pass before registry admission.	Registry implementation; duplicate/unknown tests; fixture capability matrix. Registry tests demonstrate deterministic provider selection, duplicate rejection, unknown-provider failure, and disabled Fireblocks fixture behaviour.
BASE-07	3h	Wire the seven-step gate precedence from RFP §9: (1) global wallet enablement, (2) provider enablement, (3) operation enablement, (4) environment/segment, (5) sensitive-operation kill switch, (6) provider capability, (7) idempotency/operation-state eligibility. Each gate fail-closed with deterministic blocked response. Production activation excluded.	Canonical gate precedence matrix; disabled-operation gate tests; rollback note. Matrix tests prove every required gate must pass before workflow start and that kill-switch failure is fail-closed and auditable.
BASE-08	2h	Credential request model used by Activities to call the Polity Vault REST gateway. WIS does not call Infisical directly and does not serialise provider secrets into workflow inputs, workflow history, logs, fixtures, frontend code, or provider metadata. Credential failure matrix per RFP §9.2: missing reference, stale/rotated, expired lease, gateway unavailable, provider auth rejected, redaction failure, audit correlation missing.	Vault request contract; redaction rules; secret hygiene tests. No-secret tests, credential failure matrix tests, and code scan prove credentials are requested only through the Vault gateway boundary at Activity execution time with no fallback secret path.

Stop condition (RFP §14.1): Stop if provider SDK activation, production accounts, or direct secret-store access becomes necessary. Branch guidance: feature/wallet-base/base-02-contract-registry-flags.

4.3 MR-BASE-03 - Provider-Neutral Persistence, Idempotency, Rollback (15 hours)

MR-BASE-03 covers WBS-BASE-09 through WBS-BASE-13. The package delivers the six provider-neutral tables, seed data and legacy-read compatibility, the WALLET_OPERATIONS ledger with deterministic idempotency, and the persistence rollback package. The acceptance gate is migration rollback proof, repository tests, idempotency replay tests, and state-transition tests all passing. Sensitive payload storage, private keys, provider API secrets, transaction signing data, and provider-specific backfill are explicitly out of scope.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-09	4h	Reversible gormigrate steps appended to internal/storage/migrations for providers, provider_accounts, wallets, wallet_assets, provider_metadata. Schema versioning, field-level classification notes per RFP §13.1 BASE-REQ-18, retention/redaction assumptions, and display-safe metadata. Secret fields, private keys, and signing payloads excluded.	Migration up/down scripts; schema notes; model updates. Up/down logs, schema diff, and storage tests show provider-neutral rows are reversible and contain no credential material.
BASE-10	2h	Seed provider metadata for the approved release chain set (per RFP §3.1 and Glossary). Existing DFNS/Safeheron wallet rows remain readable during Strangler Fig coexistence. Non-baseline networks rejected or disabled.	Seed rows; read-compatibility tests; chain-boundary note. Tests prove approved chain set metadata works, non-baseline networks rejected or disabled, legacy rows still read correctly.
BASE-11	4h	WALLET_OPERATIONS table created before provider side effects with deterministic idempotency key, provider-agreement evidence reference (per BASE-REQ-15), operation type, provider_id, chain, user, correlation id, state, failure class, retry count, compensation fields, and redacted evidence references. Unique constraint on (provider_id, operation_type, idempotency_key). Signing data and provider secrets excluded.	Operation ledger migration; unique constraints; model/repository stubs. Migration and repository tests prove uniqueness, deterministic replay, redacted evidence storage, and no sensitive payload persistence.
BASE-12	3h	WalletRepository methods (14 provider-agnostic methods covering the six table families per RFP §20.2). RecordOperation creates or finds WALLET_OPERATIONS before external effects and returns CONFLICT_IDEMPOTENT where appropriate. Provider calls before operation correlation are non-conforming.	Repository methods; concurrency tests; replay tests. Concurrent request tests and replay tests prove duplicate requests do not create duplicate provider side effects.
BASE-13	2h	Package migration rollback evidence, backup assumptions, residual risks, and data compatibility notes for provider-neutral persistence and the operation ledger.	Rollback note; residual-risk register; migration evidence index. Reviewer can replay up/down evidence and identify data-risk blockers before provider-specific work starts.

Stop condition (RFP §14.1): Stop if any secret-bearing or provider-specific payload is required in base persistence. Branch guidance: feature/wallet-base/base-03-persistence-idempotency.

4.4 MR-BASE-04 - Direct Temporal SDK, Worker, Workflows, Activity/Vault Boundary, Retry, Compensation (14 hours)

MR-BASE-04 covers WBS-BASE-14 through WBS-BASE-18. The package uses the direct Temporal Go SDK posture defined by RFP §8 and §20.1.8-§20.1.10, with worker bootstrap, deterministic workflow logic, Activity/Vault boundaries, retry, and SAGA compensation. The acceptance gate is workflow unit tests, local worker smoke test, history scan, Vault Activity tests, and no-Nexus proof. Provisioning Temporal Server, EKS, Helm, ArgoCD, Worker Controller, or active Nexus routing is explicitly out of scope.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-14	3h	Direct Temporal client. ExecuteWorkflow with stable workflow IDs derived from tenant, operation type, provider, target entity, and idempotency key. Namespace polity-wallet, task queue wallet-operations. Tier 1 base workflows: ProvisionWalletWorkflow and RegisterAccountWorkflow. Workflow inputs are serialisable non-secret identifiers and stable operation context only. Nexus activation excluded; wallet.nexus.routing.enabled remains false.	Workflow start code; ID/task-queue convention; no-Nexus proof. Workflow-start tests or smoke evidence prove direct invocation, task queue alignment, and operation id correlation.
BASE-15	3h	Worker endpoint at wallet-infrastructure-service/cmd/worker/main.go (or existing equivalent identified at BASE-01) registering ProvisionWalletWorkflow, RegisterAccountWorkflow, Tier 3 Canonical Activities (FetchServiceCredentials, CreateProviderAccount, StoreCredentials, CreateWallet, AddAsset, CommitProvisioning, ValidateProviderConnectivity, RevokeProviderAccount, RevokeStoredCredentials, ReconcileWallets), adapters, retry policies, and task queue configuration. Temporal Server, Helm, EKS, ArgoCD, Worker Controller provisioning, and Tier 2 Provider-Specific Procedures remain platform/provider-RFP scope.	Worker bootstrap; configuration sample; local worker smoke. Local worker proof shows the configured worker polls the expected task queue without assuming unmanaged platform deployment.
BASE-16	3h	Workflow determinism: no secrets, SDK clients, file I/O, database writes, direct Vault calls, provider clients, or non-deterministic side effects in workflow state. External calls and persistence writes belong in service/Activity boundaries.	Workflow tests; replay tests; history-scan note. Replay/history inspection proves workflow payloads contain only ids and display-safe metadata, not credentials or provider SDK objects.
BASE-17	3h	Activities fetch credentials through the Polity Vault REST gateway at Activity execution time, call the selected ProviderAdapter, map canonical errors, redact evidence, and update WALLET_OPERATIONS. Tier 3 Canonical Activities implement the ten names enumerated in RFP §11 (BASE-17) and §20.1.9. Direct Infisical calls and provider secrets in logs or history excluded.	Activity tests; Vault gateway tests; redaction evidence. Activity tests and log/history scans prove credentials are loaded only at Activity time and do not leak to persistence, logs, or workflow history.
BASE-18	2h	Map canonical retry, non-retryable, timeout, provider conflict, compensation started, compensation failed, and deferred states to the operation ledger. Explicit SAGA pattern via github.com/sunny-b/temporal-saga-go (the binding framework per RFP §20.1.10): forward-recovery, step-status tracking, idempotency keys, per-Activity compensation registration, reverse-order compensation on downstream failure.	Retry matrix; SAGA state tests; compensation evidence. Tests show retry limits, terminal state transitions, compensation fields, and evidence references are deterministic and reviewable.

Stop condition (RFP §14.1): Stop if Temporal infrastructure provisioning or active Nexus routing is required to finish the base MR. Branch guidance: feature/wallet-base/base-04-temporal-vault-boundary.

4.5 MR-BASE-05 - Unified API, Route Fate, Negative Proof, Compatibility, Callback Envelope (12 hours)

MR-BASE-05 covers WBS-BASE-19 through WBS-BASE-23. The package introduces the unified API bridge, the route fate matrix, negative proof for prohibited operation classes, Dynamic/Veriff/Lukka compatibility evidence, and the callback envelope boundary. The acceptance gate is route scan, OpenAPI/handler tests, compatibility tests, callback tests, and prohibited-operation negative proofs all passing. New provider-specific public routes, production webhook enrollment, direct provider signing, raw broadcast, payment, and WalletConnect invocation are explicitly out of scope.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-19	3h	/api/wallets/v1/* skeleton (per RFP §20.1.1 and §20.1.6) or explicit time-boxed compatibility bridge in the existing internal/rest/router and internal/rest/api packages. Accepted DFNS/Safeheron read behaviour preserved. New provider-specific public route families excluded.	Unified route skeleton; handler tests; OpenAPI or route contract note. Handler tests and response samples prove existing reads remain stable and sensitive-operation routes fail closed.
BASE-20	2h	Classify every wallet route as preserve, wrap, deprecate, disable, or remove (per RFP §11 BASE-20 and §20.9), with reason and rollback path. HTTP 301 redirection of legacy endpoints deferred to R1.3.3 per the wallet implementation baseline. Direct provider signing, raw broadcast, payment, WalletConnect, scan, and unauthorised admin paths must not remain reachable.	Route fate matrix; OpenAPI delta; rollback/deprecation note. Before/after route inventory and runtime checks prove the accepted route fate for each sensitive and legacy path.
BASE-21	3h	Static code scan with admin-scope analysis (per RFP §10.1) and runtime API inventory proving payment, signing, raw broadcast, transaction execution, WalletConnect invocation, scan, provider dashboard, webhook/callback activation, admin paths, scheduled jobs, frontend WalletConnect surfaces, and services excluded from the production build but reachable in development/proof-lane environments are removed, disabled, unreachable, dispositioned, or exception-approved. Scan scope includes admin-scope analysis. Custodian attestation with Listed Wallet Provider supporting evidence per RFP §10.1.	Sensitive-operation evidence pack; disabled-operation tests; route proof. Static scan, runtime route inventory, admin-scope analysis, provider-side machine-verifiable evidence, and negative operation tests show no unapproved custodial or transaction-execution path is reachable.
BASE-22	2h	Verify Dynamic sign-in (V3/V4), wallet linking, wallet display, Veriff/Lukka non-interference assumptions, approved release chain boundary, read-only visibility response shape, graceful degradation metadata, and display-safe API responses. Frontend redesign and Dynamic SDK/dashboard migration excluded.	Compatibility note; response samples; supported-network tests. Compatibility tests or samples show accepted auth and display flows still work and non-baseline chains stay disabled or rejected.
BASE-23	2h	Common callback envelope (per RFP §9.1 and §20.1.12) with raw-body preservation, signature verification result, provider event type, provider event	Callback envelope; ignored/deferred tests; provider handoff note. Tests prove invalid, duplicate, unknown, stale, and

WBS	Hrs	Approach	Deliverable / Evidence
		id, target operation id, redacted payload metadata, duplicate/replay treatment, and ignored-or-deferred classification. Fail-closed before mutation. Provider signatures, subscriptions, endpoint secrets, and sandbox enrollment are provider-RFP scope.	sensitive/deferred callbacks fail closed before mutation and do not activate provider runtime subscriptions.

Stop condition (RFP §14.1): Stop if new provider-specific public routes or production webhook/provider activation is required. Branch guidance: feature/wallet-base/base-05-api-compatibility-closure.

4.6 MR-BASE-06 - Traceability, Runbook, Provider Handoff, Final QA, Release-Boundary Review (7 hours)

MR-BASE-06 covers WBS-BASE-24 through WBS-BASE-26. The package closes CSV traceability, runbook, provider-start handoff, final QA, and release-boundary review. The acceptance gate is evidence index, runbook, blocked register, test/lint output or approved exceptions, and final review memo accepted. Provider runtime execution without accepted base closure or governance-approved blockers is explicitly out of scope.

WBS	Hrs	Approach	Deliverable / Evidence
BASE-24	2h	CSV-ready evidence index mapping every WBS row to FR/NFR (BASE-FR-01..07, BASE-NFR-01..06), TOR gate (AC-01..AC-05, AC-07..AC-10, G0..G7 once supplied), MR-BASE package, test evidence, rollback evidence, owner, and blocked condition. Generic non-BASE MR namespaces prohibited.	Traceability matrix; evidence index; blocked-condition register. Reviewer can trace each WBS row to requirements, MR-BASE package, evidence, and unresolved blocker without assumptions.
BASE-25	2h	Rollback/runbook material, provider-start prerequisites, open blockers, and handoff rules for Fireblocks, DFNS, and Safeheron RFPs. Provider implementation must not start unless base contract, persistence, Temporal, Vault, route, and sensitive-operation evidence are accepted or explicitly blocked.	Runbook; provider-start checklist; dependency closure note. Handoff checklist shows accepted base artefacts, remaining blockers, rollback path, and provider-specific exclusions.
BASE-26	3h	Run or document go test ./..., golangci-lint run ./..., affected polity_tests integration evidence, route scans, secret scans, and release-boundary review. Proposal acceptance does not expand provider custody, signing, raw broadcast, payment, WalletConnect, or chain scope.	QA log; release-boundary note; final review memo. Final evidence pack includes command output or approved exceptions plus reviewer sign-off that the R1.3.2 boundary is unchanged.

Stop condition (RFP §14.1): Stop if provider runtime execution is required before base closure is accepted. Branch guidance: feature/wallet-base/base-06-evidence-provider-handoff.

4.7 Package Hour Summary

Total: 67 hours, matching the RFP Effort/Budget Baseline. All hour allocations are taken directly from RFP §11 WBS without modification.

MR Package	WBS Rows	Hours	Theme
MR-BASE-01	BASE-01..03	6h	Current-state inventory and base contract freeze
MR-BASE-02	BASE-04..08	13h	Contract, registry, capability, gates, Vault boundary
MR-BASE-03	BASE-09..13	15h	Provider-neutral persistence and idempotency
MR-BASE-04	BASE-14..18	14h	Direct Temporal SDK, worker, workflows, Activity/Vault, SAGA
MR-BASE-05	BASE-19..23	12h	Unified API, route fate, negative proof, compatibility, callback envelope
MR-BASE-06	BASE-24..26	7h	Traceability, runbook, provider handoff, final QA
Total	BASE-01..26	67h	Full base scope - one fixed-price package

5. Security and Compliance Approach

Security is embedded throughout the engagement and is aligned to the non-negotiable security invariants in RFP §7-§10, including the sensitive-operation acceptance rule in §10.1.

5.1 Polity Vault REST Gateway and Secret Hygiene

- All provider credentials are requested through the Polity Vault REST gateway at Activity execution time. WIS does not call Infisical directly.
- Workflows carry only non-secret identifiers and stable operation context. No credentials, raw provider secrets, private keys, signing payloads, payment instructions, raw transactions, or provider SDK clients are passed as workflow inputs or persisted in workflow history.
- Code scans, log scans, fixture scans, frontend scans, and Temporal history scans accompany each affected MR-BASE package to prove no secret leakage.
- Credential failure matrix (per RFP §9.2) is implemented as canonical errors: missing reference, stale/rotated, expired lease, gateway unavailable, provider auth rejected, redaction failure, audit correlation missing.

5.2 Sensitive-Operation Closure (per RFP §10.1)

Sensitive-operation evidence is treated as a release-blocking acceptance artefact with three proof parts:

- Static code scan including admin-scope analysis across WIS and reachable wallet-sensitive repository surfaces (admin routes, scheduled jobs, frontend WalletConnect surfaces, services excluded from production build but reachable in dev/proof-lane), citing DSI-12-01 through DSI-12-05.
- Runtime API inventory confirming no active route accepts signing, payment, raw broadcast, transaction execution, WalletConnect invocation, custody-control payloads, or unsigned raw transaction payloads in the acceptance lane.
- Custodian attestation with supporting Listed Wallet Provider evidence where applicable, proving Polity lacks key share, signing authority, workspace key authority, and admin override authority.

Both static (compile-time capability closure) and runtime (feature gate kill switch) proofs are required. Neither alone is sufficient.

5.3 Seven-Step Gate Precedence Chain (per RFP §9)

Order	Gate	Failure result
1	Global wallet enablement	Reject before workflow start; record blocked wallet operation evidence
2	Provider enablement	Return provider unavailable or disabled; do not construct provider SDK/client
3	Operation enablement	Return unsupported or disabled operation; no provider side effect
4	Environment or segment gate	Block outside the approved local, proof-lane, ERE, or release lane
5	Sensitive-operation kill switch	Return not supported or blocked for signing, raw broadcast, payment, WalletConnect, transaction execution, scan, and sensitive admin paths
6	Provider capability	Capability metadata may confirm technical support but never authorises execution alone

Order	Gate	Failure result
7	Idempotency and op-state	Duplicate, replayed, concurrent, or already-completed operations resolve without duplicate provider side effects

5.4 Audit, Traceability, and DNO Handling

- Every state transition (provider registered, wallet provisioned, operation recorded, sensitive-operation block, capability change) writes to WALLET_OPERATIONS with redacted evidence references and is captured in the CSV-ready traceability index.
- Personal data classification and retention notes are attached to wallet tables and provider metadata fields per BASE-REQ-18 and GDPR Art. 5, 6, 28, 33.
- All RFP-derived artefacts, working notes, evidence, and proposal correspondence are handled under the RFP Internal, DNO-Gated - S3 regime: distribution to named recipients only, signed NDA in force, controlled retention, return or destruction on engagement closure.
- AI-assisted coding controls: no Polity Confidential Information, secrets, Vault references, workflow inputs/history, signing payloads, raw provider keys, or live database extracts are transmitted to any third-party AI provider. AI-produced artefacts are treated as Turntabl's own work product for licensing, warranty, and acceptance purposes, and are reviewed by a senior engineer before commit.

6. Implementation Plan and Timeline

6.1 Delivery Schedule

Sequential delivery across the six MR-BASE packages. Each package is self-contained and reversible; each package is gated on the preceding package's acceptance under the Polity Programme Lead/Product Owner sign-off process in RFP §12.2. Total elapsed duration is estimated at 4-5 weeks of business days, subject to Polity review turnaround.

Package	Hours	Timeline	Acceptance Gate (per RFP §14.1)
MR-BASE-01 Inventory + Contract Freeze	6h	T+0 to T+3	Contract freeze, route/table inventory, sensitive-path inventory, exception log accepted
MR-BASE-02 Contract + Registry + Vault Boundary	13h	T+3 to T+8	Interface tests, registry tests, disabled-operation tests, gate matrix, no-secret Vault tests pass
MR-BASE-03 Persistence + Idempotency	15h	T+8 to T+13	Migration rollback proof, repository tests, idempotency replay, state-transition tests pass
MR-BASE-04 Temporal + Worker + SAGA	14h	T+13 to T+18	Workflow tests, local worker smoke, history scan, Vault Activity tests, no-Nexus proof pass
MR-BASE-05 Unified API + Route Fate + Negative Proof	12h	T+18 to T+22	Route scan, handler/OpenAPI tests, compatibility tests, callback tests, prohibited-operation negatives pass
MR-BASE-06 Traceability + Handoff + Final QA	7h	T+22 to T+25	Evidence index, runbook, blocked register, test/lint output or approved exceptions, final review memo accepted

Timeline references (T+N) represent business days from project kickoff. Where milestone delivery is paused by Polity-provided prerequisites, review feedback, or dependency gates outside vendor control, milestone and payment dates shift on a day-for-day basis per RFP §12.1, with the pause cause recorded in the MR description or blocked-condition register.

6.2 Merge Request Strategy

Six MR-BASE packages are delivered per RFP §14, with branch guidance from §14.1 followed where the current repository branch policy and CI layout permit. Each MR references its WBS row(s), governing TOR, and vendor-facing baseline labels; includes a stop-condition note; carries the evidence focus described in RFP §14.1; and passes CI/CD checks before submission. WBS rows may share an MR within the same MR-BASE namespace where the shared review purpose remains coherent; oversized MRs carry a written rationale tied to migration atomicity, Temporal boundary integrity, or sensitive-operation closure (per BASE-NFR-05).

6.3 Team Structure

A lean, senior team of two engineers is proposed for this backend-only Go engagement under the AI-assisted coding productivity baseline:

Role	Responsibilities	WBS Coverage
Technical Lead	Architecture alignment, Temporal/SAGA pattern design, security review, code review, Polity baseline traceability, Polity Architecture liaison, sensitive-operation closure evidence, provider handoff documentation	Oversight of MR-BASE-01..06; primary on BASE-03, BASE-05, BASE-07, BASE-18, BASE-21, BASE-24..26
Senior Backend Engineer (Go)	Inventory, schema migrations, WalletRepository, Temporal worker bootstrap and workflows, Tier 3 Canonical Activities, API handlers, callback envelope, feature gate wiring, CSV traceability index	Primary on BASE-01..02, BASE-04, BASE-06, BASE-08..17, BASE-19..20, BASE-22..23

Both engineers have production experience with Go, GORM/gormigrate, and centralised secret-management gateways. Technical Lead provides code review on every MR and is the primary point of contact for the named Polity contact (Julian Weber, IO).

7. Risk Management and Quality Assurance

7.1 Risk Register (extended from RFP §16)

Risk	Severity	Our Control	Rollback or Escalation
R-01: Polity baseline conflict	HIGH	Treat conflicts as PDK §11.1 change-control items. Stop implementation outside approved scope until the §11.1 decision is recorded. Raise written exception to the Approval Authority with requirement and architecture delta.	Hold affected MR before merge until §11.1 decision recorded
R-02: Internal sensitive-operation baseline breach	HIGH	Three-part sensitive-operation proof (static + runtime + custodian attestation), runtime route inventory with admin-scope analysis, feature gates on sensitive paths, fail-closed kill switch.	Disable affected routes; revert to last approved provider-specific behaviour
R-03: Provider API or sandbox instability	MEDIUM	Provider sandboxes used where available; mock-adapter tests as the primary acceptance path; explicit dependency dates and blocked-evidence register.	Defer affected provider runtime feature while preserving base adapter contract
R-04: Migration data loss	MEDIUM	Reversible gormigrate steps with tested Rollback functions; backup assumptions captured in BASE-13 rollback package; CI runs forward+backward and diffs schema.	Restore backup; keep legacy read path active until acceptance
R-05: Parallel RFP divergence	MEDIUM	Base adapter contract frozen at BASE-03 before any provider RFP starts; written exception required for divergence.	Reject non-conforming proposals; hold provider work at stub integration
R-06: IP and code ownership	MEDIUM	Bespoke work product assigned to Polity per the resulting contract; open-source dependencies and licences disclosed; Polity-specific implementation patterns not reused in non-Polity engagements without written consent.	Vendor remediates licensing or assignment defects before closure
R-07: Data residency and DNO handling	MEDIUM	All RFP material routed through Polity-controlled channels; signed NDA in force; controlled storage on team devices; destruction or return on engagement closure.	Attestation of destruction; escalation to governance if breach suspected
R-08: Provider sandbox cascade unavailability	MEDIUM	Mock adapters and recorded fixtures used as primary acceptance path; live sandbox treated as additive; blocked-evidence register per provider.	Accept fixture-based evidence with governance-approved exception and live-sandbox closure item

Risk	Severity	Our Control	Rollback or Escalation
R-09: Frontend cutover dependency deadlock	MEDIUM	Apply read-only visibility carve-out in RFP §5.2 where sensitive-operation closure permits; backend-ready state tracked separately from frontend-routed state.	Hold frontend cutover at legacy-safe path; close Base RFP on backend-ready evidence
R-10: Evidence reproducibility	MEDIUM	Artefact capture (logs, screenshots, exported workflow histories, scan output) included in evidence pack rather than reference to live state; live-only evidence treated as supplementary.	Withhold acceptance until captured artefacts available; treat irreproducible evidence as blocked-condition entry
R-11: Dynamic V3/V4 auth dependency drift	MEDIUM	Dynamic V3/V4 dependency state recorded at WBS-BASE-01 discovery; any user-visible Dynamic regression treated as base-acceptance blocker; acceptance windows coordinated with FE/Auth owner.	Pause base-RFP acceptance; revert to last known-good Dynamic configuration
R-12: Currency and cross-border vendor risk	LOW	EUR quotation per RFP Quotation Currency metadata; no FX exposure passed to Polity; standard EU/Swiss tax treatment assumed and confirmed at contract.	Re-quotation under stated currency and risk allocation rules if assumptions change
R-13: AI-assisted coding controls	MEDIUM	No Polity Confidential Information, secrets, Vault references, workflow inputs/history, signing payloads, raw provider keys, or live database extracts transmitted to any third-party AI provider. All AI output reviewed by senior engineer before commit. AI-produced code treated as Turntabl's own work product for licensing, warranty, acceptance.	Confirmed transmission treated as security event; withhold acceptance until provenance re-established

7.2 Quality Assurance Approach

Test plan follows the RFP §17 proposal requirement (item 5) and the QA Checklist in RFP §15. Coverage spans:

- Unit tests (MR-BASE-02..05): ProviderAdapter, ProviderCredentials, AdapterRegistry, ProviderCapabilities, canonical errors, disabled-provider fixtures, capability matrix.
- Integration tests (MR-BASE-03..05): migration up/down on a test Postgres, WalletRepository methods, idempotency replay, WALLET_OPERATIONS state transitions, route inventory before/after, callback envelope fail-closed paths.
- Migration evidence (MR-BASE-03): forward+backward migration logs, schema diff, seed/read compatibility for DFNS/Safeheron rows, rollback proof.
- Route inventory (MR-BASE-01, MR-BASE-05): static route scan and runtime API inventory across internal/rest/router and internal/rest/api, with sensitive-path classification.
- Dynamic compatibility (MR-BASE-05 WBS-BASE-22): sign-in, wallet linking, wallet display non-regression; Veriff/Lukka non-interference; approved release chain boundary tests.
- Sensitive-operation negative-capability evidence (MR-BASE-05 WBS-BASE-21): static scan with admin-scope analysis, runtime API inventory, custodian attestation with Listed Wallet Provider supporting evidence where applicable.

- Temporal workflow tests (MR-BASE-04): direct ExecuteWorkflow proof, deterministic replay, history scan for secret leakage, Activity unit tests, local worker smoke, Nexus-disabled proof.
- Final QA (MR-BASE-06): go test ./..., golangci-lint run ./..., affected polity_tests integration evidence, secret scans, release-boundary review.

8. Assumptions and Dependencies

8.1 Named Assumptions (per RFP §17 item 3)

- Turntabl has read/write access to the wallet-infrastructure-service Go repository from project kickoff, including internal/rest/router/router.go, internal/rest/api/api.go, internal/service, internal/storage/migrations, internal/adapters/{dfns,safeheron,fireblocks}, providers/polityVault, and config/config.go.
- A staging Postgres instance is available for migration round-trip testing, with permissions to CREATE and DROP tables. Postgres version to be confirmed at WBS-BASE-01 inventory and assumed compatible with the existing gormigrate history.
- Polity Vault REST gateway endpoint, authentication mechanism, and credential reference shape are documented and available at engagement start. WIS will not call Infisical directly.
- A Temporal Cloud or self-hosted Temporal server is accessible from the staging environment, with connection string, polity-wallet namespace, and wallet-operations task queue config supplied. Local Docker Compose Temporal is used where the EKS/proof-lane handoff is incomplete, with the dependency recorded in the blocked-condition register.
- AI-assisted coding is accepted as the delivery mode. The 67-hour Effort/Budget Baseline is achievable under this assumption.
- The Polity-named contact (Julian Weber, IO) is available for technical clarifications and MR review within 3 business days per package.
- AC-01..AC-10 and G0..G7 TOR section references are supplied at engagement start under PDK §11.1 amendment-approval authority (per RFP §6.2), as are PDK identifiers for the Polity baselines listed in RFP §6.1.
- Provider sandbox credentials for DFNS and Safeheron are available (or mock adapters are accepted as primary evidence path) for the affected MR-BASE packages.
- No scope changes or baseline modifications occur during execution outside the formal PDK §11.1 change-control process.

8.2 Exclusions (per RFP §17 item 3)

- Provider runtime activation (Fireblocks SDK lifecycle, DFNS key ceremony, Safeheron MPC setup) - assigned to provider-specific RFPs after base acceptance.
- Dynamic.xyz SDK or dashboard migration - FE/Auth scope unless separately assigned.
- Frontend redesign, UniApp/SuperApp Cabinet UI changes, Fireblocks Web SDK bridge - separate scope.
- Infrastructure provisioning (Terraform, Helm, EKS, ArgoCD, Worker Controller, OpenTofu, Infisical Operator) - consumed from the Polity infrastructure delivery baseline.
- Production webhook enrollment, provider sandbox credential issuance, legal/provider account setup - provider/legal scope.
- HTTP 301 redirection of legacy endpoints - deferred to R1.3.3 per the wallet implementation baseline.
- Nexus runtime invocation - disabled in R1.3.2; wallet.nexus.routing.enabled remains false.
- Sensitive-operation activation (signing, payment, raw broadcast, transaction execution, WalletConnect, scan, custody-control payloads, sensitive admin paths) - blocked unless a separately approved governance release authorises them.

8.3 Dependencies

Dependency	Requirement	Owner
wallet-infrastructure-service repo	Read/write access from project kickoff	Polity DevOps
Staging Postgres database	Postgres instance with CREATE/DROP table permissions	Polity DevOps
Polity Vault REST gateway	Endpoint, auth mechanism, credential reference shape documented at kickoff	Polity Programme
Temporal server	Connection string, polity-wallet namespace, wallet-operations task queue	Polity Infrastructure
temporal-saga-go	github.com/sunny-b/temporal-saga-go available via Go module proxy	Turntabl
TOR / PDK identifiers	AC-01..AC-10, G0..G7 TOR section refs; PDK ids for baselines per RFP §6.1	Polity Programme
Provider sandbox / mock adapters	DFNS and Safeheron sandbox access or accepted mock-adapter evidence	Polity / Turntabl
polity_tests repository	Identifier and access path for affected integration evidence	Polity Programme
Polity Architecture review	Sign-off on BASE-03 contract freeze and BASE-20 route fate matrix	Polity Architecture
Provider-RFP team contact	One member available to review provider-start handoff in MR-BASE-06	Polity PMO

8.4 Conflicts With Polity Baselines (per RFP §17 item 6)

At the time of proposal submission, Turntabl has identified no material conflict between this proposal and the Polity baselines enumerated in RFP §6.1 (Polity system requirements baseline, Polity system architecture baseline, Polity wallet architecture baseline, Polity wallet implementation baseline, Polity release and evidence baseline, Polity governance baseline, PDK/DRS/NTS/DFG/CSV standards). Should the WBS-BASE-01 current-state inventory surface a conflict, it will be raised as a written exception under PDK §11.1 before implementation of the affected WBS row proceeds.

9. Commercial Proposal

9.1 Pricing Structure

This proposal is submitted on a Fixed-Price basis for the total scope of work defined across WBS-BASE-01 through WBS-BASE-26 (one base fixed-price package per RFP §12.1). Pricing is quoted in EUR (€) per the RFP Quotation Currency metadata row, exclusive of VAT or equivalent local tax, with tax handled separately at the resulting contract.

Role	Hours	Blended Rate (EUR/hr)	Subtotal (EUR)
Technical Lead	17h	€42	€714
Senior Backend Engineer (Go)	50h	€42	€2,100
Total	67h	€42	€2,814

- AI Enterprise Tooling (upgrade from Standard to Premium, 2 months): €174
- Total Fixed Price: €2,988 (exclusive of VAT)
- Hour allocation by MR-BASE package matches RFP §11 WBS exactly: MR-BASE-01 6h / MR-BASE-02 13h / MR-BASE-03 15h / MR-BASE-04 14h / MR-BASE-05 12h / MR-BASE-06 7h. The 67-hour total accepts the AI-assisted coding productivity baseline declared in the RFP Delivery Methodology Assumption row.
- Contingency: separately shown at 15% of fixed price (€410 / approximately 10 hours), approval-only, used only after written Polity approval per RFP §12.1. Not pre-approved.
- Rates are inclusive of vendor overhead and profit. Remote delivery; no travel expenses. No additional licensing required (temporal-saga-go is open source).

9.2 Payment Schedule (per RFP §12.2)

Six-milestone payment schedule, each milestone contingent on written acceptance by the Polity Programme Lead/Product Owner or delegated IO and the mapped technical reviewer:

%	Milestone	Trigger	Amount (EUR)
10%	Project Kick-off	Executed contract signature, NDA in force, access/onboarding confirmation	€
10%	Discovery Acceptance	Written acceptance of MR-BASE-01 (WBS-BASE-01-03) with no unresolved blocking findings	€
20%	Mid-Term Milestone	Written acceptance of the first coherent MR-BASE package set representing roughly 50% of WBS effort and evidence	€
30%	Final QA Submission	Vendor delivery of completed evidence pack and written certification of WBS completion, ready for Polity review	€
25%	Final Acceptance	Polity-issued written Final Acceptance memo following review and closure of accepted findings	€

%	Milestone	Trigger	Amount (EUR)
5%	Warranty Release	90 calendar days after Final Acceptance, subject to no material defect raised during the warranty window	€
100%	Total		€

9.3 Service Credits and Delay Treatment (per RFP §12.1)

Where a milestone is delayed by a cause attributable to Turntabl (and not by Polity-provided prerequisites, review feedback, or dependency gates), a service credit of zero point five percent (0.5%) of the fixed price accrues per full week of delay beyond the accepted milestone date, up to a cumulative cap of five percent (5%) of fixed price (€2,814). Service credits are applied as a deduction against the next due milestone payment as an agreed pre-estimate of loss for delay; they are not a penalty and do not limit other Polity remedies for material breach.

Where milestone delivery is paused by Polity-provided prerequisites, review feedback, or dependency gates outside Turntabl's control, the corresponding milestone date and payment date shift on a day-for-day basis. No service credit accrues during such pauses. The pause cause is recorded in the MR description or blocked-condition register and execution resumes promptly on removal of the cause.

9.4 Warranty Holdback (per RFP §12.1)

Five percent (5%) of the fixed price (€2,814) is retained as a Warranty Release milestone, payable 90 calendar days after Final Acceptance subject to no material defect having been raised by Polity during the warranty window. Where a material defect is raised, the warranty window is paused until the defect is remediated to Polity's written satisfaction, after which the remainder of the 90-day window resumes.

9.5 Change Control and Contingency Use

Any work outside WBS-BASE-01 through WBS-BASE-26, deferred work, production activation, provider-specific runtime work, or sensitive-operation expansion requires a written change request approved before execution per RFP §12.1. Use of the 15% contingency requires written Polity approval with documented justification (early notification to the Programme Manager, detailed cause, impact). Overrun against Turntabl's quoted hours is Turntabl's risk; overrun against Polity's 67-hour reference is not, by itself, grounds for a change request.

9.6 Optional Post-Delivery Support (Separate Line Item)

As a separately priced line item (not part of the fixed-price scope), Turntabl offers post-delivery support for the first 30 calendar days after Final Acceptance at a rate of €45 per hour, billed on a time-and-materials basis. This covers bug fixes in base-layer code, clarifications required by downstream provider RFP teams, and minor adjustments discovered during provider adapter development. This is optional and does not duplicate the warranty holdback.

9.7 Budget Assumptions

- All rates are in EUR, exclusive of VAT or equivalent local tax, inclusive of vendor overhead and profit.
- Remote delivery; no travel expenses. Polity provides all necessary infrastructure, repository access, staging environment, Temporal server (or proof-lane handoff), and Polity Vault REST gateway access.

- No additional licensing fees expected for the base scope. AI-assisted coding tooling licensed by Turntabl at Turntabl's cost.
- The 67-hour estimate is achievable based on the AI-assisted coding productivity baseline. If WBS-BASE-01 inventory surfaces materially different repository conditions, Turntabl will notify Polity immediately and propose use of the 15% contingency with written justification rather than absorbing hidden work.